Project Development Phase

Coding & Solution

Date	03 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented code and the overall OptiCrop solution, demonstrating working functionality, clean code practices, and alignment with the defined requirements.

Solution Summary

Field	**Details**
Repository Link / URL	https://github.com/vijayapadmapriya/OptiCrop
Programming Language(s)	Python 3.8+
Framework(s) Used	Streamlit (frontend/web app), scikit-learn (ML model), pandas & numpy (data handling)
Key Features Implemented	7-parameter soil/climate input form; Random Forest crop prediction; confidence score display; top-3 alternate crop suggestions; input validation; error handling
Pending / Incomplete Features	None for the core use case. Planned future enhancements: multi-language support, live weather API integration, mobile-friendly layout (see Scalability & Future Plan document)
Setup / Run Instructions	pip install -r requirements.txt, then streamlit run app.py — see README.md and Project Executable Files document for full steps

Code Quality Checklist

S.No	**Criteria**	**Status (Yes/No)**

1	Code is modular and organized into functions / classes	Partial — training and inference are split into separate files (train_model.py, app.py); app.py itself is mostly a linear Streamlit script
2	Meaningful variable and function names are used	Yes — N, P, K, temperature, humidity, ph, rainfall match the domain directly
3	Code includes comments / documentation where necessary	Yes — train_model.py has step-by-step comments; app.py has a header comment with run instructions
4	Error handling is implemented for critical operations	Yes — model loading and prediction are wrapped in try/except with user-facing messages
5	The application runs without critical errors	Yes — tested locally: model loads, predicts, and returns results correctly
6	Code is committed to a version control repository	Yes — pushed to GitHub (see Repository Link above)

Additional Notes / Comments

The trained model (model.pkl) achieves 99% accuracy on the held-out test split of the Crop Recommendation dataset (2200 records, 22 crop classes, sourced from Kaggle). The dataset (Crop_recommendation.csv) and training script (train_model.py) are both included in the repository so the model can be reproduced or retrained from scratch.